

Project report

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STUDENT MANAGEMENT SYSTEM

CRUD-Based Web Application Development

1. Introduction

The **Student Management System** is a CRUD-based web application developed to manage student records efficiently. The system provides a simple interface for registering students and managing their information.

The application allows the user to perform the four fundamental database operations:

- **Create** – Add a new student record.
- **Read** – View existing student records.
- **Update** – Modify student information.
- **Delete** – Remove student records.

The project uses a modern web interface connected to a **Supabase PostgreSQL database**. The application was developed with the support of **Lovable AI** and uses a React-based frontend.

2. Problem Statement

Managing student information manually can be time-consuming and may result in errors or difficulty in maintaining records.

The Student Management System provides a digital solution where student information can be stored, searched, updated, and deleted

easily. The system also provides a dashboard to display basic information such as the total number of students and departments.

3. Objectives

The main objectives of the project are:

- 1. To develop a web-based student management application.
 - 2. To implement CRUD operations.
 - 3. To store student information in a PostgreSQL database.
 - 4. To connect the frontend application with Supabase.
 - 5. To provide search functionality for student records.
 - 6. To provide edit and delete operations.
 - 7. To validate student information.
 - 8. To provide a simple and user-friendly interface.
 - 9. To gain practical knowledge of full-stack web application development.
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4. Technology Stack

Technol ogy	Purpose
Lovable AI	Application development assistance
React	Frontend development
HTML	Web structure
CSS	User interface styling
JavaScri pt	Application functionality

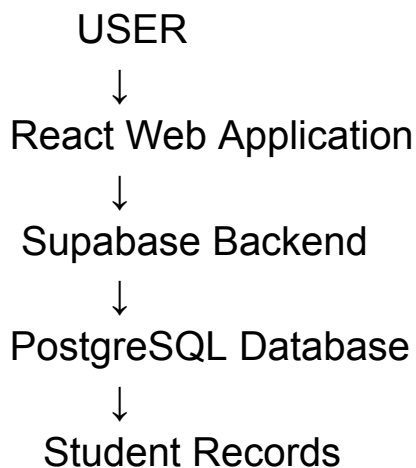
Supabase Backend and database services

PostgreSQL Data storage

Git/GitHub Source code management

5. System Architecture

The application follows a frontend–backend–database architecture.



The user interacts with the React-based interface. The application communicates with Supabase to perform database operations. Student information is permanently stored in the PostgreSQL database.

6. Database Design

A table named **students** was created in the Supabase PostgreSQL database.

Students Table

Field Name	Data Type	Description
id	int8	Primary key and unique record ID
student_id	text	Student identification number
name	text	Student name
department	text	Student department
email	text	Student email address
phone	text	Student phone number
year_of_study	int4	Student year of study

The **id** field is used as the primary key. The database contains student records that are displayed in the application.

7. User Interface

The Student Management System provides a clean and simple dashboard.

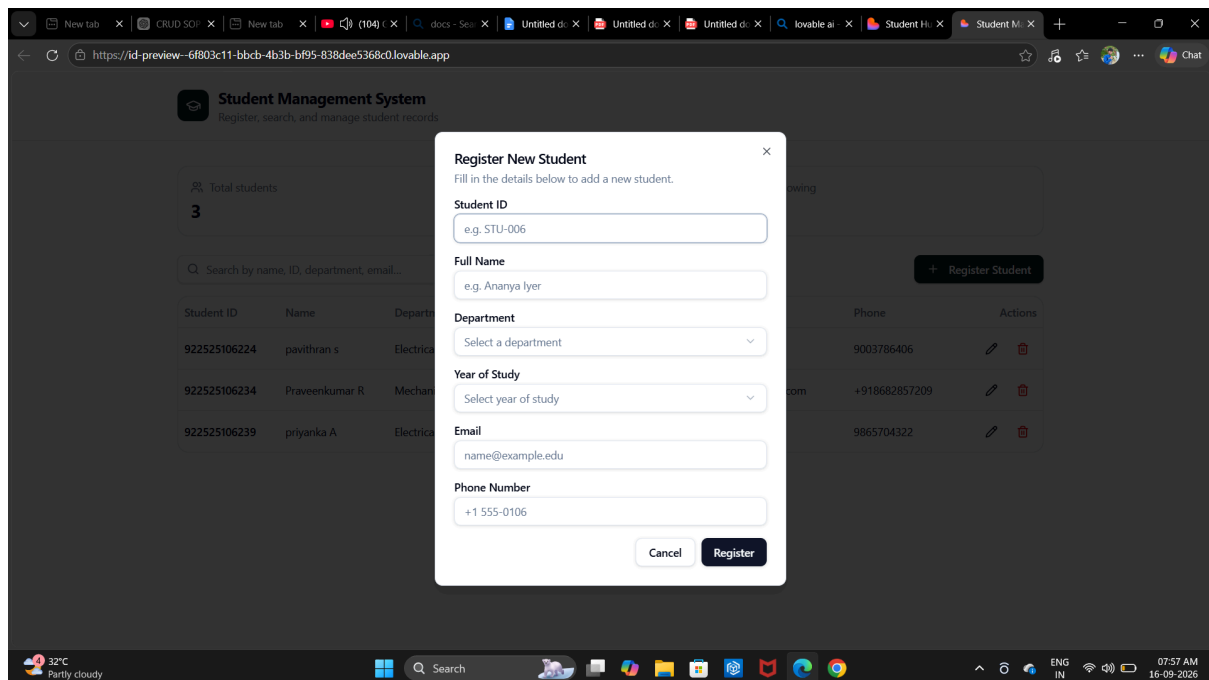
The main dashboard contains:

- Student Management System title
- Total students count
- Number of departments
- Number of displayed records
- Search bar
- Register Student button

- Student records table
- Edit action
- Delete action

In the completed application shown in the screenshot, **3 student records** are displayed and students belong to **2 departments**.

8. Student Registration



The **Register Student** option is used to add a new student to the system.

The registration form collects information such as:

- Student ID
- Student Name
- Department
- Year of Study
- Email
- Phone Number

After entering valid information and submitting the form, the student record is stored in the Supabase PostgreSQL database.

9. CRUD Operations

9.1 Create

The Create operation is used to add a new student.

Process:

Enter Student Details



Submit Registration Form



Validate Data



Insert into Supabase



Display Student in Table

9.2 Read

The Read operation retrieves student records from the database and displays them in the application.

The dashboard displays the student ID, name, department, year of study, email, and phone number.

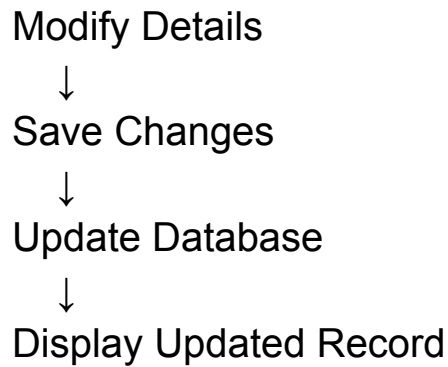
9.3 Update

The Edit icon available in the **Actions** column is used to modify an existing student record.

Process:

Select Edit



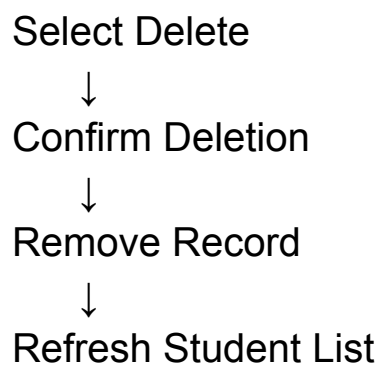


9.4 Delete

The Delete icon is used to remove a student record.

The system can request confirmation before permanently deleting the selected record.

Process:



10. Search Functionality

The application provides a search box with the option to search student information.

The search can be performed using information such as:

- Student name
- Student ID
- Department
- Email

This allows the user to find a particular student quickly without manually checking every record.

11. Database Integration

The application is integrated with **Supabase**.

The Supabase project contains a PostgreSQL table named **students**. The screenshot of the Supabase Table Editor confirms that student records are stored in the database.

The database currently contains three records, including:

- Praveenkumar R
- Pavithran S
- Priyanka A

The database table contains the corresponding student IDs, departments, email addresses, and phone numbers.

12. Testing

The application was tested by performing the main CRUD operations.

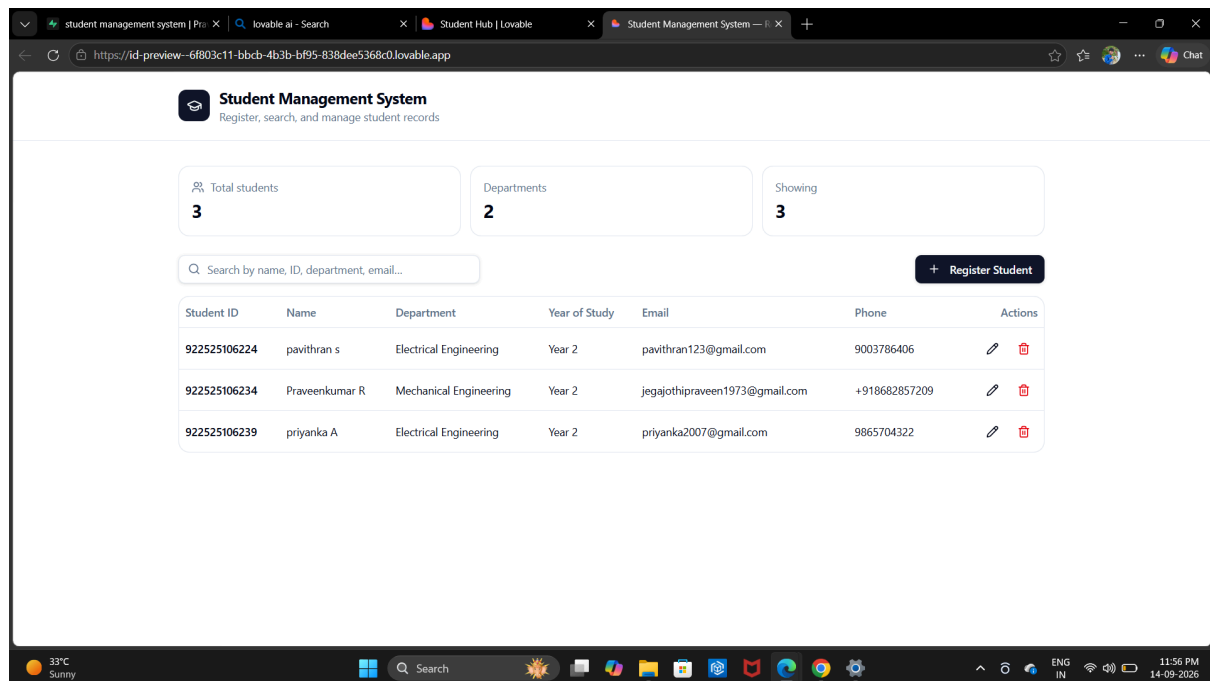
Test Case	Operation	Expected Result	Result
TC01	Add student	New record created	Pass
TC02	View students	Records displayed	Pass
TC03	Edit student	Record updated	Pass
TC04	Delete student	Record removed	Pass

TC05	Search	Matching records displayed	Pass
TC06	Database verification	Data stored in Supabase	Pass

The application interface and database were verified after adding student records.

13. Screenshots

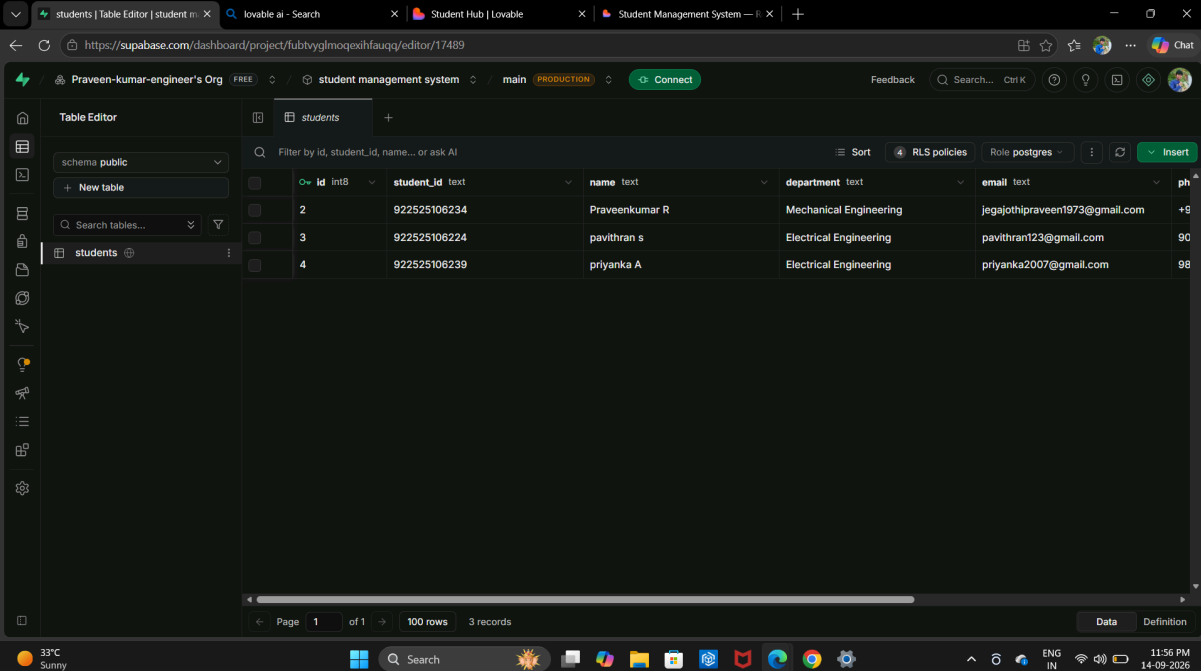
Figure 1: Student Management System Dashboard



Description:

The screenshot shows the main dashboard of the Student Management System. It displays the total number of students, number of departments, search functionality, Register Student button, and the student records table. Three student records are displayed in the application.

Figure 2: Supabase PostgreSQL Database



The screenshot displays the Supabase Table Editor interface for a table named 'students'. The table is located in the 'public' schema. The columns are: 'id' (integer), 'student_id' (text), 'name' (text), 'department' (text), 'email' (text), and 'ph' (text). There are 3 records in the table. The interface includes a sidebar with navigation options, a search bar, and a 'Connect' button. The bottom status bar shows the system time as 11:56 PM on 14-09-2026.

id	student_id	name	department	email	ph
2	922525106234	Praveenkumar R	Mechanical Engineering	jegajothi.praveen1973@gmail.com	+6
3	922525106224	pavithran s	Electrical Engineering	pavithran123@gmail.com	90
4	922525106239	priyanka A	Electrical Engineering	priyanka2007@gmail.com	98

Description:

The screenshot shows the `students` table in the Supabase Table Editor. The table contains student records stored in the PostgreSQL database. The columns include `id`, `student_id`, `name`, `department`, `email`, and other student information.

14. Advantages

- Easy to use.
- Reduces manual record management.
- Provides quick student searching.
- Supports complete CRUD operations.
- Stores data in a PostgreSQL database.
- Provides a responsive web interface.

- Makes student record management faster and more organized.
 - Provides practical experience in full-stack development.
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15. Challenges and Solutions

Challenge 1: Database Integration

Connecting the web application to the database was an important part of the project.

Solution:

Supabase was integrated with the application to provide database connectivity and PostgreSQL data storage.

Challenge 2: CRUD Implementation

Implementing Create, Read, Update, and Delete operations required proper communication between the frontend and database.

Solution:

The application was connected to the Supabase database and each CRUD operation was tested individually.

Challenge 3: Data Management

Student records needed to be displayed clearly and managed efficiently.

Solution:

A dashboard with a structured student table, search functionality, and Edit/Delete actions was implemented.

16. Future Enhancements

The following features can be added in future versions:

1. Student login and authentication.
 2. Admin dashboard.
 3. Department-wise student reports.
 4. Advanced filtering.
 5. Export student records to Excel/PDF.
 6. Student profile pages.
 7. Attendance management.
 8. Marks and academic performance management.
 9. Email notifications.
 10. Deployment with a custom domain.
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17. Learning Outcomes

Through this project, I gained practical knowledge of:

- Web application development.
 - React-based frontend development.
 - CRUD operations.
 - PostgreSQL database management.
 - Supabase integration.
 - Frontend–backend communication.
 - Form validation.
 - Search and filtering.
 - Database testing.
 - Software project documentation.
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18. Conclusion

The **Student Management System** was successfully developed as a CRUD-based web application. The system provides functionality

to create, read, update, and delete student records through a user-friendly interface.

The application was successfully connected to a **Supabase PostgreSQL database**, and student records were stored and displayed through the web interface. This project provided practical experience in developing and integrating a complete database-driven web application.